Quantitative Risk Assessment of Market Volatility Under Macroeconomic Impacts

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Introduction:

Financial markets are increasingly driven by macroeconomic shocks (e.g., CPI releases, FOMC rate decisions). Traditional risk models often assume asset returns follow a Normal distribution, leading to a significant underestimation of "tail risk" during market stress.

This project develops an end-to-end interactive dashboard using Python to bridge the gap between raw market data and advanced risk econometrics. The objective is to:

1. Automate the collection of market data and mapping of macroeconomic events.
2. Validate statistical properties (Stationarity, Normality).
3. Model volatility using **GARCH(1,1)** with a **t-distribution**.
4. Compute and backtest **Value-at-Risk (VaR)** to quantify potential portfolio losses.

Methods & Procedures:

The analysis pipeline was implemented in Python using the following approach:

- **Data Engineering:**
  - Retrieved adjusted close prices (2022–2024) via `yfinance`.
  - Mapped non-trading day macro events (BLS/FOMC data) to the nearest subsequent trading day to analyze immediate market impact.
- **Statistical Validation:**
  - **Stationarity:** Validated log-returns using the Augmented Dickey-Fuller (ADF) test.
  - **Normality:** Applied Shapiro-Wilk and Jarque-Bera tests to detect skewness and excess kurtosis (fat tails).
- **Modeling:**
  - **ARIMA(1,1,0):** modeled the linear trend of log-prices.
  - **GARCH(1,1):** modeled conditional volatility to account for leptokurtic behavior (fat tails) in returns.
- **Risk Metrics & Backtesting:**
  - Calculated **95% Value-at-Risk (VaR)** based on dynamic volatility forecasts.

Results:

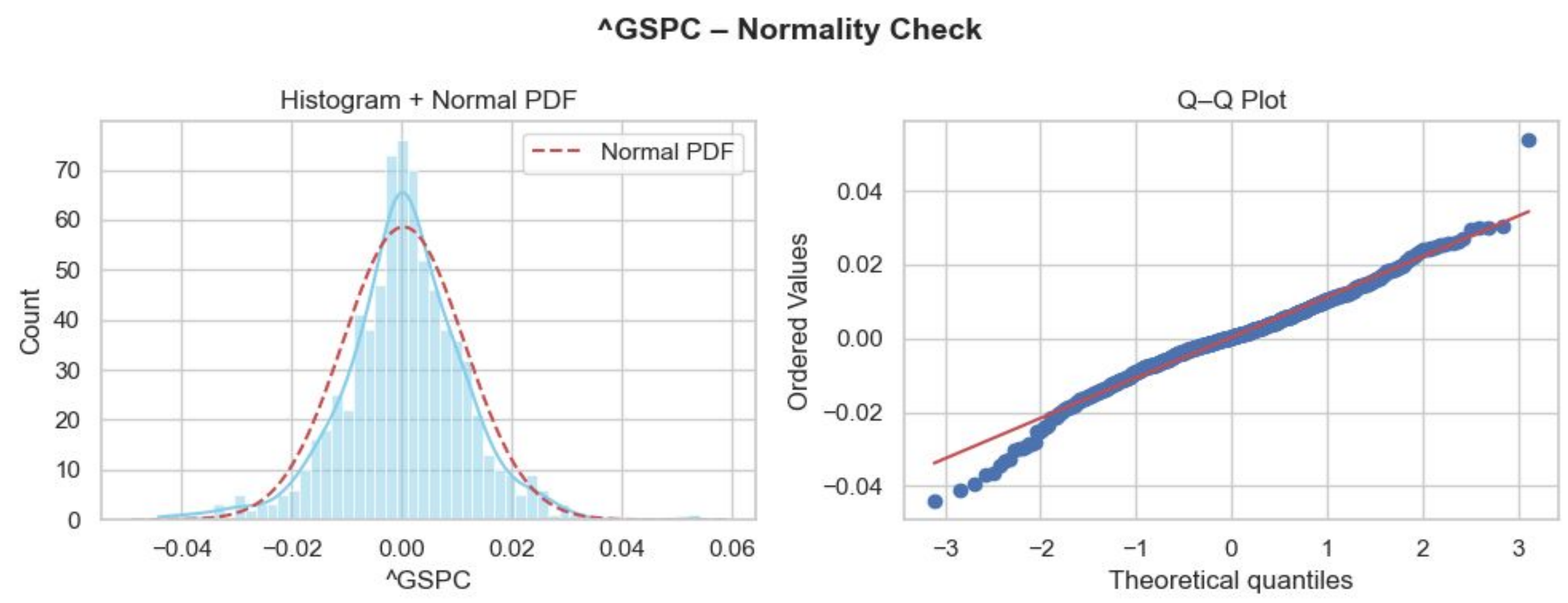

Figure 1: S&P 500 Normality Check. The Histogram (left) shows a peak higher than the Normal PDF. The Q-Q Plot (right) shows data points diverging from the red line at the extremes, confirming "fat tails."

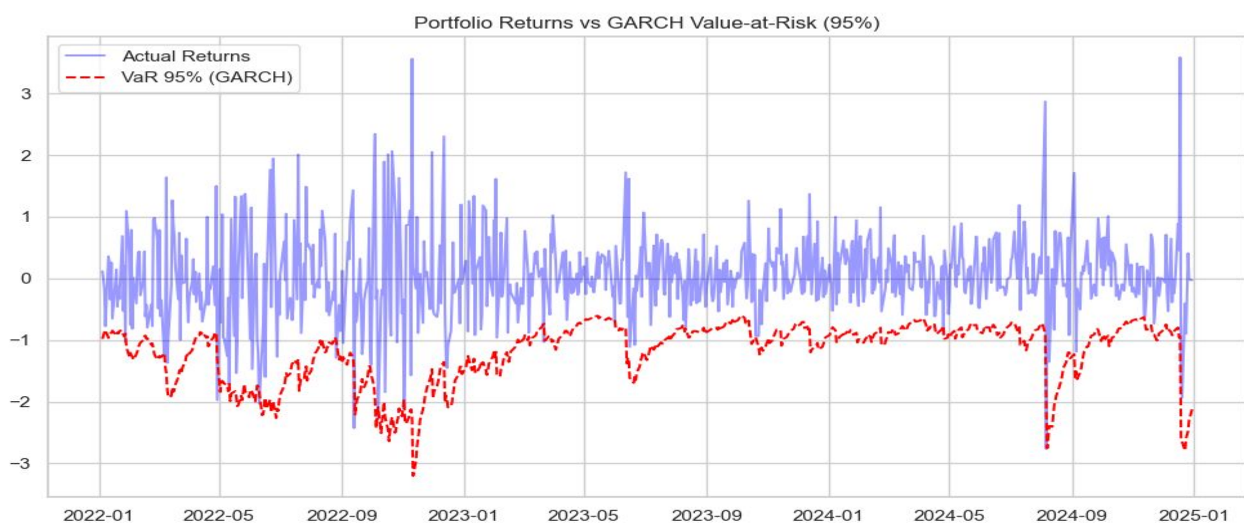

Figure 2: Portfolio Returns vs. 95% GARCH VaR. The red dashed line (VaR) dynamically adjusts to market volatility. Red dots indicate days where actual losses exceeded the model's prediction.

Analysis:

1. Non-Normality Confirmed: Figure 1, the S&P 500 log-returns deviate significantly from the theoretical Normal distribution (red line). The heavy tails observed in the Q-Q plot justify the use of the Student-t distribution in the GARCH model rather than a standard Gaussian approach.

2. Dynamic Risk Modeling: Figure 2 demonstrates that the GARCH model successfully adapts to volatility clusters. During stable periods, the VaR tightens; during stress periods (e.g., 2022 rate hikes), the VaR expands to provide a wider safety buffer.

3. Model Accuracy: The backtesting results (Kupiec Test) yielded a p-value > 0.05, indicating we "Fail to Reject" the model. This confirms that the GARCH-based VaR is statistically valid for this time period.

Conclusions:

This project successfully demonstrates a reproducible workflow for Quantitative Risk Management. By moving beyond static variance assumptions and adopting dynamic volatility modeling (GARCH), investors can better anticipate downside risk. The results show that accounting for "fat tails" is essential for accurate risk estimation in modern financial markets.

Future Work:

- **Implement Asymmetric Volatility Models:** Extend the GARCH(1,1) framework to asymmetric specifications (e.g., EGARCH, GJR-GARCH) to capture the leverage effect, where negative shocks impact volatility more strongly than positive shocks.
- **Adoption of Expected Shortfall (ES):** Complement Value-at-Risk (VaR) with Expected Shortfall (Conditional VaR) to better estimate the average loss exceeding the 95% threshold, addressing the "fat tail" risks identified in the Q-Q plots.

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